Matrices & Matrix Algebra

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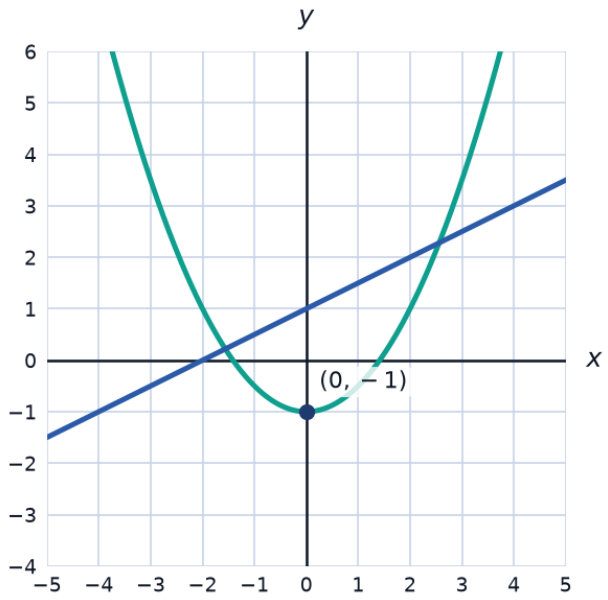
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Matrix Multiplication Properties**

Find the standard matrix $[T]$ of the linear operator $T : \mathbb{R}^2 \to \mathbb{R}^2$ that performs a horizontal shear by factor 2, followed by a counterclockwise rotation of 45°.

2 **[Foundational]** • **Target: Matrix Inverses and Elementary Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 3, 4 \rangle$.



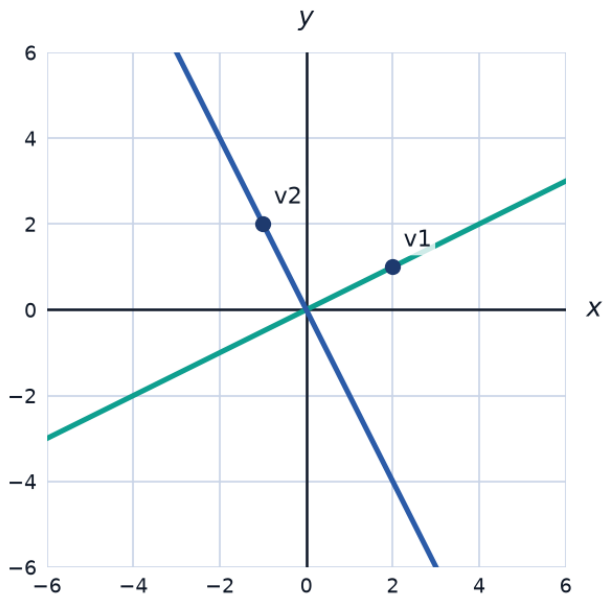
3 **[Foundational]** • **Target: Symmetric, Orthogonal, and Special Matrices**

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix $A = (4, 1; 2, 3)$. Find the matrix representation $[T]_{\mathcal{B}}$ relative to the basis $\mathcal{B} = \{(1; 1), (-1; 2)\}$ via similarity transformation $P^{-1}AP$.

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4 [Foundational] • Target: Linear Transformations and Standard Matrices

Use the Gauss–Jordan elimination algorithm on the augmented block matrix $[A \mid I_3]$ to compute A^{-1} for: $A = (1, 0, 2; 2, -1, 3; 4, 1, 8)$.

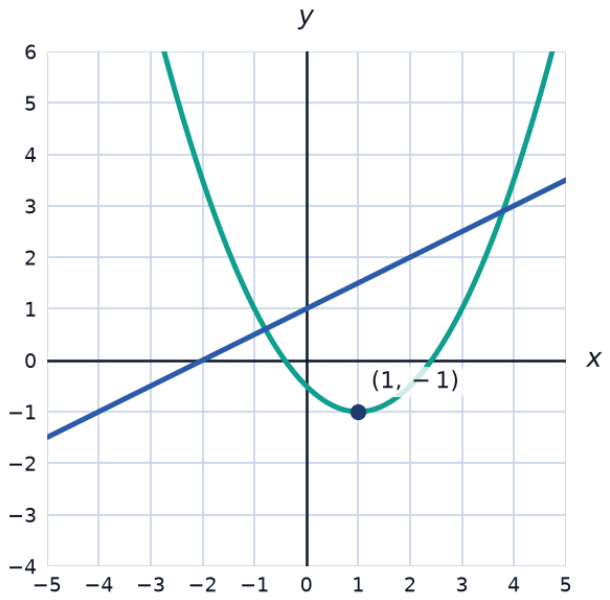


5 [Foundational] • Target: Matrix Multiplication Properties

Consider a matrix $A = (2, -1; 3, k)$. If the determinant of the matrix A^2 is 100, find the positive value of the parameter k.

6 [Foundational] • Target: Matrix Inverses and Elementary Matrices

Prove that if λ is an eigenvalue of an invertible matrix A, then $\lambda eq 0$ and λ^{-1} is an eigenvalue of A^{-1} with the exact same corresponding eigenvector.



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7 [Foundational] • Target: Symmetric, Orthogonal, and Special Matrices

Diagonalize the matrix $A = (1, 2; 2, 1)$ by finding invertible P and diagonal D such that $A = PDP^{-1}$. Then compute A^k for any positive integer k.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Linear Transformations and Standard Matrices

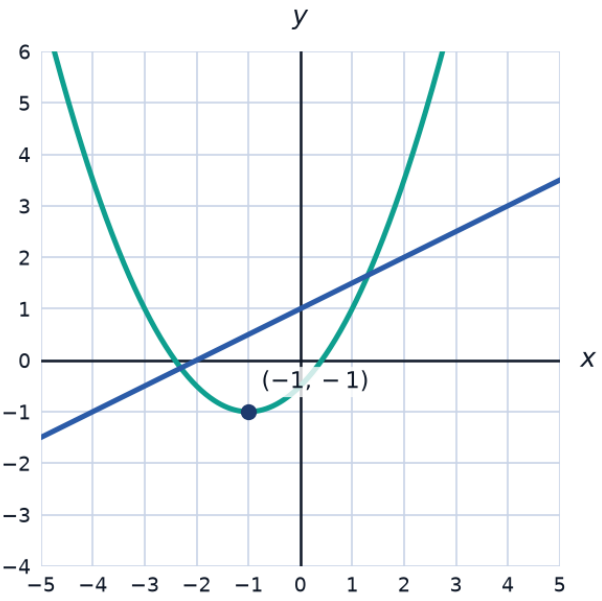
For the matrix $A = (1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5)$, find bases for $\text{Row}(A)$, $\text{Col}(A)$, and $\text{Null}(A)$. Verify the Rank–Nullity Theorem.

9 [Intermediate] • Target: Matrix Multiplication Properties

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = (1; 2)$. If the trace of A is 7, find the other eigenvalue of A.

10 [Intermediate] • Target: Matrix Inverses and Elementary Matrices

Let $A = (1, 2, 0; 0, 3, 1; 2, 1, 4)$. Compute $\det(A)$, find the adjugate matrix $\text{adj}(A)$, and verify the identity $A\text{adj}(A) = \det(A)I_3$.



11 [Intermediate] • Target: Symmetric, Orthogonal, and Special Matrices

Let A and B be invertible $n \times n$ matrices. Prove algebraically that: (a) $(AB)^T = B^T A^T$, and (b) $(AB)^{-1} = B^{-1}A^{-1}$.

12 [Intermediate] • Target: Linear Transformations and Standard Matrices

Use Cramer's Rule to solve for x_2 in the linear system:

$\{2x_1 + x_2 + x_3 = 4; x_1 - x_2 + 2x_3 = 1; 3x_1 + 2x_2 - x_3 = 7$

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13 [Intermediate] • Target: Matrix Multiplication Properties

Compute the determinant of the 4×4 matrix by Laplace cofactor expansion along an optimal row or column: $A = (2, 0, -1, 3; 0, 5, 0, 0; 1, -2, 3, 1; 4, 0, 1, 2)$.

14 [Intermediate] • Target: Matrix Inverses and Elementary Matrices

Consider the matrix $A = (2, 1; 1, 3)$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $(x'; y') = A(x; y)$, what is the image of the point $(3, 1)$ under this transformation?

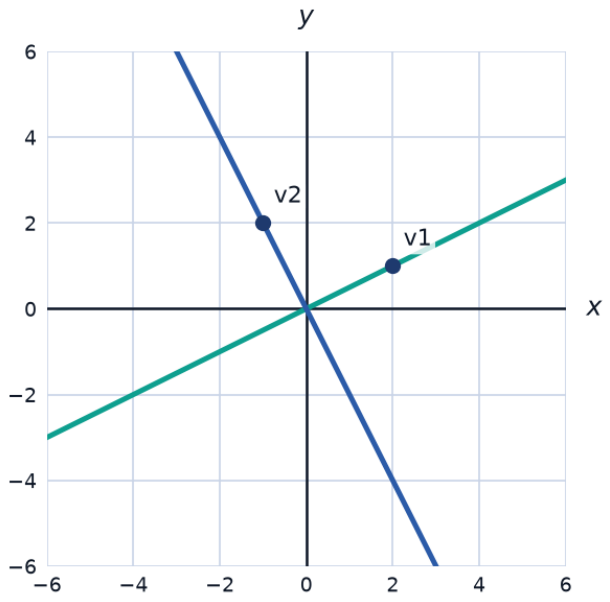
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the component orthogonal to $\mathbf{u}$ of $\mathbf{y} = \langle -2, 3, -4 \rangle$ for $\mathbf{u} = \langle 1, -3, 3 \rangle$.

16 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 4.01 Determinants

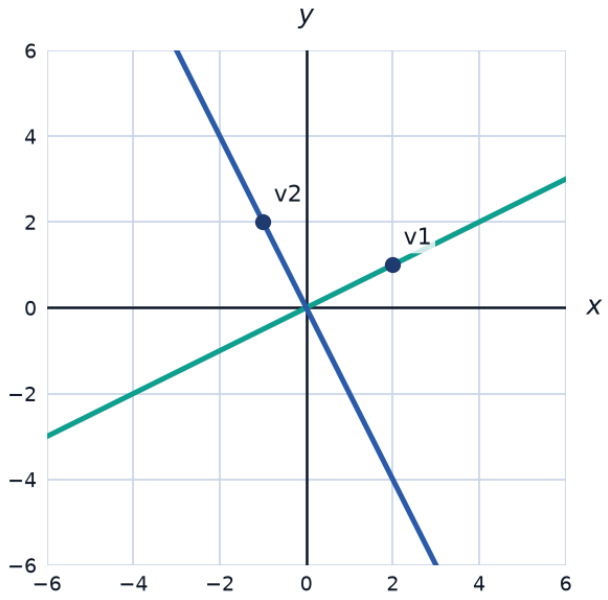
Prove that if $\mathbf{x}_p$ is any particular solution to $A\mathbf{x} = \mathbf{b}$, then the complete solution set is the affine subspace $S = \{\mathbf{x}_p + \mathbf{x}_h : \mathbf{x}_h \in \text{Null}(A)\}$.



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17 [Challenging] • Target: Matrix Multiplication Properties • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the shortest Euclidean distance from the point $P(3, -1, 4)$ to the hyperplane $\mathcal{H}: 2x - y + 2z = 5$ in $\mathbb{R}^3$.



18 [Challenging] • Target: Matrix Inverses and Elementary Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace for: $A = (3, 1, 0; 0, 3, 0; 0, 0, 2)$. Is A diagonalizable?

19 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 4.01 Determinants

Apply the Gram–Schmidt orthogonalization process to the columns of A to find its QR factorization: $A = (1, 2; 2, 1; -2, 2)$.

20 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Perform a complete bifurcation analysis of the system in terms of parameter $k \in \mathbb{R}$: $\{x + y + kz = 2; 3x + 4y + 2z = k; 2x + 3y - z = 1$ Determine all values of k for which the system has (a) a unique solution, (b) infinitely many solutions, and (c) no solution.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Matrix Multiplication Properties

If matrix A has size 3×4 and matrix B has size 4×2, what is the size of the product AB?

a. 3×2

b. 4×4

c. 2×3

d. Multiplication is undefined

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22 [Foundational] • Target: Matrix Multiplication Properties

Is matrix multiplication commutative in general ($AB = BA$)?

a. No, $AB \neq BA$ in general even when both products are defined

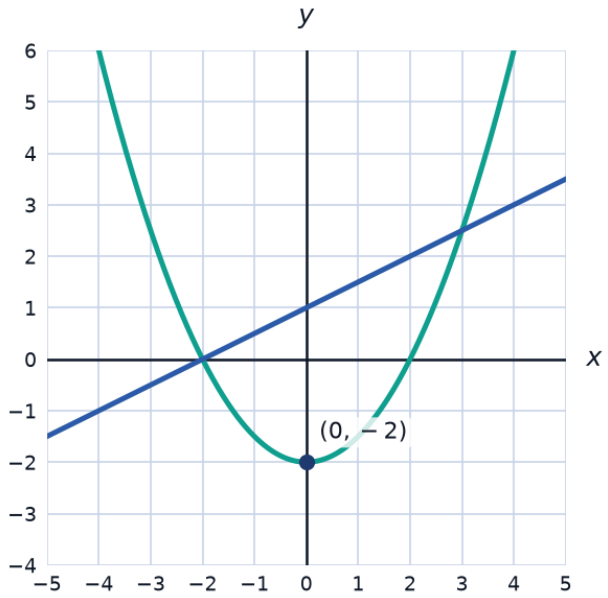
b. Yes, multiplication of numbers and matrices is always commutative

c. Only if the entries are integers

d. Yes, provided both matrices are square

23 [Foundational] • Target: Matrix Multiplication Properties

What is $(AB)^T$ equal to in terms of the transposes A^T and B^T?



a. $B^T A^T$

b. $A^T B^T$

c. AB^T

d. BA^T

24 [Foundational] • Target: Matrix Inverses and Elementary Matrices

Find the inverse of the 2×2 matrix $A = [[4, 7], [1, 2]]$.

a. $[[2, -7], [-1, 4]]$

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

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25 [Foundational] • Target: Matrix Inverses and Elementary Matrices

What is an elementary matrix?

a. A matrix obtained by performing a single elementary row operation on an identity matrix

b. Any matrix containing only 0s and 1s

c. A diagonal matrix with determinant 1

d. A matrix with exactly one non-zero row

26 [Foundational] • Target: Matrix Inverses and Elementary Matrices

If A, B, C are invertible square matrices, what is $(ABC)^{-1}$?

a. $C^{-1}B^{-1}A^{-1}$

b. $A^{-1}B^{-1}C^{-1}$

c. CBA

d. ABC^{-1}

27 [Foundational] • Target: Symmetric, Orthogonal, and Special Matrices

A square matrix A is symmetric if and only if:

a. $A^T = A$

b. $A^T = -A$

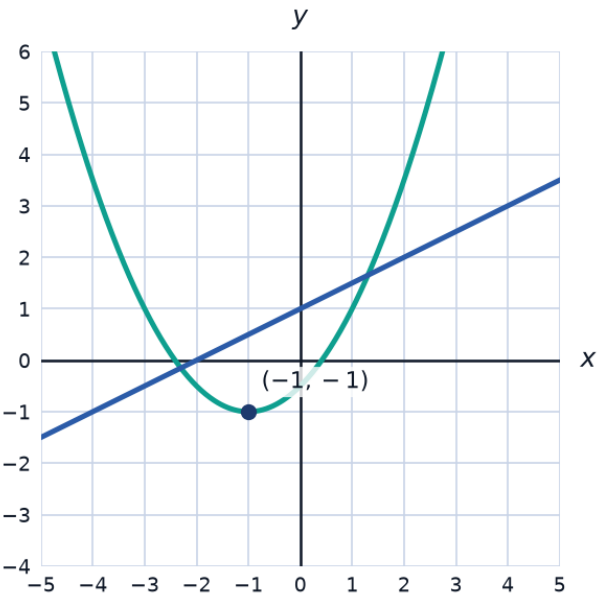
c. $A^T = A^{-1}$

d. $\det(A) = 1$

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28 [Foundational] • Target: Symmetric, Orthogonal, and Special Matrices

What defines an orthogonal matrix Q?



a. $Q^TQ = I$, meaning $Q^{-1} = Q^T$

b. $Q^2 = I$

c. $\det(Q) = 0$

d. All entries of Q are perpendicular

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Symmetric, Orthogonal, and Special Matrices

If Q is an orthogonal matrix, what are the possible values for $\det(Q)$?

a. ± 1

b. 0

c. 1 only

d. Any real number

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30 [Intermediate] • Target: Linear Transformations and Standard Matrices

A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c:

a. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$

b. $T(\mathbf{u} \cdot \mathbf{v}) = T(\mathbf{u}) \cdot T(\mathbf{v})$

c. $T(\mathbf{u}) = ||\mathbf{u}||$

d. $T(\mathbf{0}) = [1, 1, \ldots, 1]^T$

31 [Intermediate] • Target: Linear Transformations and Standard Matrices

What is the standard matrix for counterclockwise rotation in $\mathbb{R}^2$ by angle θ?

a. $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$

b. $[[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$

c. $[[\sin\theta, \cos\theta], [\cos\theta, -\sin\theta]]$

d. $[[\cos\theta, 0], [0, \sin\theta]]$

32 [Intermediate] • Target: Linear Transformations and Standard Matrices

What is the standard matrix for reflection across the x-axis in $\mathbb{R}^2$?

a. $[[1, 0], [0, -1]]$

b. $[[-1, 0], [0, 1]]$

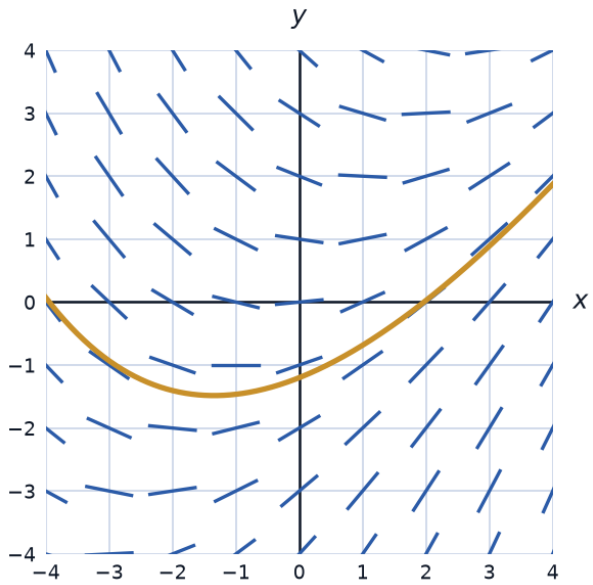
c. $[[0, 1], [1, 0]]$

d. $[[-1, 0], [0, -1]]$

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33 [Intermediate] • Target: Matrix Multiplication Properties

Consider a linear transformation represented by the matrix $A = (2, 1; 1, -2)$. A point P is mapped to P' by this transformation. If the coordinates of P are $(x; y)$ and the coordinates of P' are $(x'; y')$, then $(x'; y') = A(x; y)$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.



a. 1

b. 2

c. 3

d. 4

34 [Intermediate] • Target: Matrix Inverses and Elementary Matrices

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, -1 \rangle$.

a. $[(1, -1; -1, 1)]$

b. $\left[\left(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right)\right]$

c. $[(0, -1; -1, 0)]$

d. $\left[\left(\frac{1}{2}, -\frac{1}{2}; -\frac{1}{2}, \frac{1}{2}\right)\right]$

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35 **[Intermediate]** • **Target: Symmetric, Orthogonal, and Special Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 2, -1 \rangle$.

a. $[(4, -2; -2, 1)]$

b. $\left[\left(\frac{1}{5}, \frac{2}{5}; \frac{2}{5}, \frac{4}{5}\right)\right]$

c. $\left[\left(\frac{3}{5}, -\frac{4}{5}; -\frac{4}{5}, -\frac{3}{5}\right)\right]$

d. $\left[\left(\frac{4}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{1}{5}\right)\right]$

36 **[Intermediate]** • **Target: Linear Transformations and Standard Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 3 \rangle$.

a. $[(1, 3; 3, 9)]$

b. $\left[\left(\frac{9}{10}, -\frac{3}{10}; -\frac{3}{10}, \frac{1}{10}\right)\right]$

c. $\left[\left(-\frac{4}{5}, \frac{3}{5}; \frac{3}{5}, \frac{4}{5}\right)\right]$

d. $\left[\left(\frac{1}{10}, \frac{3}{10}; \frac{3}{10}, \frac{9}{10}\right)\right]$

37 **[Intermediate]** • **Target: Matrix Multiplication Properties**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 1 \rangle$.

a. $[(1, 1; 1, 1)]$

b. $\left[\left(\frac{1}{2}, -\frac{1}{2}; -\frac{1}{2}, \frac{1}{2}\right)\right]$

c. $[(0, 1; 1, 0)]$

d. $\left[\left(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right)\right]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Matrix Inverses and Elementary Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 2\rangle$.

a. $\left[\left(\frac{1}{5}, \frac{2}{5}; \frac{2}{5}, \frac{4}{5}\right)\right]$

b. $[(1, 2; 2, 4)]$

c. $\left[\left(\frac{4}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{1}{5}\right)\right]$

d. $\left[\left(-\frac{3}{5}, \frac{4}{5}; \frac{4}{5}, \frac{3}{5}\right)\right]$

39 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 4.01 Determinants

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 2, 1\rangle$.

a. $[(4, 2; 2, 1)]$

b. $\left[\left(\frac{1}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{4}{5}\right)\right]$

c. $\left[\left(\frac{4}{5}, \frac{2}{5}; \frac{2}{5}, \frac{1}{5}\right)\right]$

d. $\left[\left(\frac{3}{5}, \frac{4}{5}; \frac{4}{5}, -\frac{3}{5}\right)\right]$

40 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Consider the matrix $A = (k, 2; 1, -3)$. If the matrix A is invertible and its inverse A^{-1} has a determinant of $-\frac{1}{11}$, find the value of k.

a. 4

b. 5

c. 3

d. 6

41 [Challenging] • Target: Matrix Multiplication Properties • Cross-Lesson: 2.01 Systems of Linear Equations

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3, -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

a. 5

b. 6

c. 4

d. 7

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42 [Challenging] • Target: Matrix Inverses and Elementary Matrices • Cross-Lesson: 4.01 Determinants

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis $\mathcal{B}$ is $[T]_{\mathcal{B}} = (a, b\ c, d)$, and $a + d = 5$, determine the value of $\det(T)$.

a. 7

b. 5

c. 3

d. 2

43 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Let A be a 2×2 matrix given by $A = (2, k; 3, -1)$. If $A^2 = (1, -7; 3, 4)$, find the value of k.

a. 2

b. 3

c. 4

d. 5

44 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (2, k\ 1, 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

a. 1

b. 2

c. 3

d. 4

45 [Challenging] • Target: Matrix Multiplication Properties • Cross-Lesson: 4.01 Determinants

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$. The area of the transformed square is known to be k. Find the value of k.

a. 5

b. 10

c. 15

d. 20

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